

Jose Roberto Tello Ayala

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EDUCATION

- **PhD Candidate in Applied Mathematics** September 2022 - May 2027 (Expected)
Harvard University Cambridge MA, USA
 - Fourth-year PhD candidate advised by Professor Finale Doshi-Velez. Part of the Data to Actionable Knowledge Lab at the Harvard John A. Paulson School of Engineering and Applied Sciences, the Fahed Lab at the Cardiovascular Disease Initiative (CVDi) of the Broad Institute of Harvard and MIT, and the Cardiovascular Research Center of Massachusetts General Hospital (MGH).
- **Master of Science in Applied Mathematics** September 2022 - May 2025
Harvard University Cambridge MA, USA
 - Awarded en route to Ph.D.
 - Relevant courses: Advanced Computer Vision, Quantitative Methods for NLP (taken at MIT), Optimal Transport for ML, Probabilistic ML, Geometric Methods for ML, Foundations of Deep Learning.
- **Bachelor of Science in Applied Mathematics** August 2015 - December 2019
Instituto Tecnológico Autónomo de México Mexico City, Mexico
 - Graduated with honors by acquiring a special mention thanks to my thesis titled: "A study concerning the Chebyshev Center problem in finite-dimensional normed vector spaces" under the supervision of Dr. César García-García.

EXPERIENCE

- **Graduate Research Assistant** September 2022 - Present
Harvard University Boston, USA
 - My research focuses on improving the interpretability of computer vision models, particularly in healthcare applications. My work centers on developing interpretable model architectures to generate meaningful insights from data, with a particular focus on healthcare applications. My work aims to improve clinical decision-making, extract relevant phenotypes from cardiac imaging in cardiovascular disease, and identify the factors driving antidepressant treatment dropout in precision psychiatry.
- **Business Intelligence Manager** September 2021 - June 2022
Maker Media Group Cuernavaca Morelos, Mexico
 - Pioneered the Business Intelligence division at Maker. Implemented machine learning, optimization, and data science solutions to improve Out-Of-Home advertisement selection utilizing geo-referenced data.
- **Research Assistant** June 2021 - January 2022
San Luis Potosí Institute of Scientific Research and Technology Remote
 - I worked for the project Test and Contain, a project originated at the University of Oxford and the San Luis Potosí Institute of Scientific Research and Technology in Mexico, under guidance of Dr. Francisco Marmolejo-Cosío. I worked on designing mechanisms for testing and safely opening Mexican universities during the COVID-19 pandemic, based on multi-objective optimization, reinforcement learning, and information theory approaches.

PUBLICATIONS

- * **Jose Roberto Tello Ayala**, Akl C. Fahed, Weiwei Pan, Eugene V. Pomerantsev, Patrick Thomas Ellinor, Anthony Philippakis, Finale Doshi-Velez **Signature Activation: A Sparse Signal View for Holistic Saliency**. Presented at ICML 3rd Workshop on Interpretable Machine Learning in Healthcare (IMLH). The paper was awarded the First Prize for Best Paper at IMLH.
- * **Jose Roberto Tello Ayala**, Kelvin Supriami, Siddharth Swaroop, Samuel F. Friedman, Ozan Unlu, Mahnaz Maddah, Roukoz Abou-Karam, Eugene Pomerantsev, Patrick T. Ellinor, Finale Doshi-Velez, Akl C. Fahed (Under review). **A Machine Learning Driven Approach to Quantifying Coronary Artery Tortuosity**.
- * Akl C. Fahed*, **Jose Roberto Tello Ayala***, Samuel Friedman*, Ozan Unlu, Marcus D. R. Klarqvist, Paolo Di Achille, Mustafa Nasir-Moin, Eugene Pomerantsev, Mahnaz Maddah, Finale Doshi-Velez, Patrick T. Ellinor, Anthony Philippakis (In preparation). **CorAI: Artificial Intelligence Based Detection of Coronary Artery Stenosis in more than 40,000 Coronary Angiograms**

- * Kelvin Supriami, Sarah M. Urbut, **José R. Tello-Ayala**, Ozan Unlu, Samuel F. Friedman, Roukoz Abou-Karam, Satoshi Koyama, Md Mesbah Uddin, Eugene Pomerantsev, Michael T. Lu, Michael C. Honigberg, Krishna G. Aragam, Finale Doshi-Velez, Aniruddh P. Patel, Pradeep Natarajan, Patrick T. Ellinor, Akl C. Fahed **Genomic Drivers of Coronary Artery Disease and Risk of Future Outcomes After Coronary Angiography** JAMA Network Open. 2025; doi:10.1001/jamanetworkopen.2024.55368
- * Inko Bovenzi*, Adi Carmel*, Michael Hu*, Rebecca Hurwitz*, Fiona McBride*, Leo Benac, **José Roberto Tello Ayala**, Finale Doshi-Velez. **Pruning the Path to Optimal Care: Identifying Systematically Suboptimal Medical Decision-Making with Inverse Reinforcement Learning** AMIA Annual Symposium proceedings. AMIA Symposium, 2024, 202–211.
- * K Supriami, S M Urbut, **J R Tello-Ayala**, O Unlu, S N Friedman, R Abou-Karam, E Pomerantsev, M T Lu, A A Philippakis, A P Patel, P Natarajan, P T Ellinor, A C Fahed **Coronary artery disease polygenic risk score in patients undergoing coronary angiography and risk of future revascularization**. European Heart Journal, Volume 45, Issue Supplement 1, October 2024, doi.org/10.1093/eurheartj/ehae666.1333
- * Luis Benavides-Vázquez, Héctor Alonso Guzmán-Gutiérrez, Jakob Jonnerby, Philip Lazos, Edwin Lock, Francisco J. Marmolejo-Cossío, Ninad Rajgopal, and **José Roberto Tello-Ayala** **Optimal Testing and Containment Strategies for Universities in Mexico amid COVID-19**. Presented at the inaugural Association for Computing Machinery (ACM) conference on Equity and Access in Algorithms, Mechanisms and Optimization (EAAMO) 2021. Article No.: 12, Pages 1 - 9 doi.org/10.1145/3465416.3483300.

HONORS AND AWARDS

- **Best Paper Award** July 2023
ICML's Interpretable Machine Learning in Healthcare Workshop
 - First place for the best paper award for our paper **Signature Activation: A Sparse Signal View for Holistic Saliency** at ICML 3rd Workshop on Interpretable Machine Learning in Healthcare (IMLH) (\$1,000).
- **First Year PhD Fellowship** 2023
 - Jerry A. and Adi K. Greenberg Fellowship for the Academic Year 2022-2023
- **Undergraduate Dissertation Special Mention** 2020
 - Special Mention for my undergraduate dissertation (2020).
- **60% Scholarship For my Undergraduate Studies** 2015
Instituto Tecnológico Autónomo de México